

CHARLIE DUVA

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OBJECTIVE

I am a highly motivated high school senior, with a passion for mathematics and statistics, who is currently pursuing an associate degree in computer science. I am seeking an internship to explore the application of computer science principles and new mathematical concepts to solve real problems under the mentorship of talented professionals in these fields.

EDUCATION

- Liberty High School, Eldersburg, Maryland
 - Carroll Community College, Westminster, Maryland
 - Computer Science Major Associate Degree
- Graduation: June 2027
Graduation: May 2027
GPA: 4.0 (4.70)

RELEVANT COURSEWORK

- Dual Enrollment Multivariable Calculus
- Dual Enrollment Linear Algebra
- AP Statistics (5)
- AP Calculus AB / BC (5)
- Dual Enrollment Discrete Structures
- Dual Enrollment Computer Science 1
- Dual Enrollment Java Programming
- AP Computer Science Principles (4)
- AP Physics 1
- AP Chemistry

SKILLS

- Java
- Python
- JavaScript
- MATLAB
- Microsoft & Google Office Suites

LEADERSHIP

- FIRST Robotics Competition (FRC) Robotics Team Captain
- FIRST Lego League Youth Mentor
- Robotics Summer Camp Counselor
- Founder & Lead of Adaptive Toys Project
- LHS Chess Team Captain and LHS Chess Club President
- Principal's Advisory Board (2023-2025)

RECOGNITION

- Carson Scholar (7 consecutive years)
- FIRST Leadership Award Semi-Finalist 2026
- Bausch + Lomb Honorary Science Award
- R.I.T. Computing Award
- National Honors Society (April 2025 – Present)
- Phi Theta Kappa Honor Society (September 2025 – Present)

EXPERIENCE

FIRST Robotics Competition (FRC) Team 2199 Team Captain/Strategy & Scouting Lead (June 2022 – Present)

- Strategic Design: Applied statistics to determine the optimal strategy for robot design and competition performance
- Scouting/Data Collection: Designed a UI that is intuitive and efficient for team data collection
- Data Analysis: Utilized competition data and developed formulas to assess teams and provide insights for alliance strategy
- Teaching a summer strategic analysis and design class for the team
- FIRST Lego League (FLL) Team Mentor (October 2025-present): Mentored 2 FLL teams by guiding them through mission strategy and innovation project development
- Programming team member: Autonomous routine
- Community Impact: Edited the annual impact essay and authored the executive summary
- Won FIRST Chesapeake District Championships in 2023, District FIRST Impact Award in 2024, 2025, & 2026

Johns Hopkins University Applied Physics Laboratory ASPIRE Internship (June 2026 – August 2026)

- Applying Bayesian search theory to model the movement of platforms using SONAR

Northrop Grumman High School Involvement Partnership Mentoring Program (November 2025 – Present)

- Work on a year-long STEM research and engineering project under the mentorship of an employee

Johns Hopkins University Applied Physics Laboratory STEM Academy (September 2025 – May 2026)

- MATLAB 1 Programming Course + MATLAB 2 Programming Course

Chess Coach (October 2023 – May 2024)

- Provided private lessons and free, monthly chess lessons at my local library
- Started the chess club at the public library and developed instruction on rules and strategy

COMMUNITY SERVICE ACTIVITIES (1000+ HOURS)**Robotics (June 2022 – Present)**

- Organized/hosted community programs to promote STEM/robotics, including science fair demonstrations, STEM summer camps, library programs, Girl Scout badge workshops, robot pumpkin smashes, and electronics recycling programs
- Started and led an adapted toys project to provide free toys and accessibility switches to kids with physical disabilities. Worked with Carroll County Public Schools adaptive technology lead to deliver toys in person
- Lead scorekeeper at our FLL competition, Roar of the Robots
- Camp counselor at our summer camps, Robot University and Jr. Robotics Academy. Taught programming, provided engineering challenges, and helped with preparation of materials and lessons

Teen Advisory Board (October 2020 – Present)

- Collaborated with teens and librarians to develop and execute programs for children at the library
- Carroll County Public Library Summer Reading Program Volunteer, 2023-2025
- Brainstormed activity ideas, created and organized necessary materials, did event set-up and clean up for lock-ins, game nights, escape rooms, video game competitions, STEM activities using Sphero robots and drones, Harry Potter celebrations, and a Renaissance Fair
- Engaged in community service projects such as answering kids' letters to Santa, designing artwork to decorate the library, and making blankets and toys for the dogs and cats at the local animal shelter

Freedom Optimist Shooting Stars Soccer Clinic Volunteer Spring/Fall 2022-2026

- Assisted special needs players ages 5-18 in learning soccer skills
- Modified the activities to fit the needs and interests of each child participant

Camp Invention Leadership Intern Program July 2021-2024

- Provided setup and cleanup of each station
- Guided lessons and crafts for campers on the topics of bioluminescence and space exploration

Leo Club October 2023 - Present

- Helped with fundraisers, such as fruit sales and carnival food stands, to finance service projects
- Participated in and helped setup the annual Wreaths Across America to honor war veterans
- Collected hats, gloves, and toiletries for the needy and pet supplies for the animal shelter

ADDITIONAL EXTRACURRICULAR ACTIVITIES

- Academic Challenge (Trivia) Team (October 2023 – Present)
- Freedom Optimist Recreational Soccer Team (September 2012 – Present)
- Piano Lessons (January 2016 – Present)